Analyzing the Success of Corner Kicks in the English Premier League

Championship Group 4:

Hayden Andrews, Lauren Havenstein, Tyler Katowitz, Daniel Larson, Gavin Rowland

1. Introduction to the Data

1.1 Introduction and Background

Soccer is a sport notoriously vacant of goals. In the English Premier League, the most-watched soccer league in the world, an average of 2.87 goals were scored in each game (Premier League, 2025). Soccer involves 11 players on each team who face off for two 45-minute halves on a field about 120 yards long and 80 yards wide. They spend the time attempting to get a ball into the opposing team's goal, a frame 8 feet tall and 24 feet wide defended by a goalkeeper, one player per team that can use their hands (Soccer.com, 2025).

As goals in soccer are very difficult to score, Corners represent one of the few structured, repeatable scoring opportunities, which is why analysts and coaches increasingly study their patterns. A corner kick occurs when the defending team last touches the ball before it crosses its own end line. The attacking team then restarts play from the corner of the field, usually by crossing the ball into a crowded penalty area. The penalty area is an 18-by-44-yard box in front of either goal where the defending goalkeeper can use their hands, and within the penalty box is a 6-yard box that protrudes only 6 yards from the goal line and is 6 yards wider than the goalposts.

1.2 Motivation and Purpose

A corner kick is a rare instance where, after the ball crosses the line adjacent to the opposing goal, the attacking team can cross the ball unopposed in what is generally called a set play. Premier League games average just over 10 corners a game, which led to 13.9% of goals, and 15.1% of shots last season (Ali, 2023). A broader study across multiple European leagues found that a team that scores a goal from a corner wins or draws 75% of the time (Plakias et al., 2025). The 20 Premier League teams scored between 4 and 13 goals directly following corners, and one team, Tottenham Hotspur, earned an additional 11 points from these goals (Win = 3 points, Tie = 1 point, Loss = 0 points). Losing or gaining 11 points could result in a 5-place change in the league table, which has significant consequences not only for winning the league, but also for prize money, qualification for European tournaments, and relegation. However, over the last three seasons, only 2.3% of corners have led to a goal, meaning that the vast majority of these 10 glorious chances each game are squandered (Wei et al., 2025). This has been noted by some teams, as 11 English Premier League teams now employ a set-piece coach, focused solely on the success of corners and other set plays (Creative Set Plays, 2025).

The purpose of this study is to analyze corner-kick effectiveness in the Premier League by identifying which measurable factors strongly predict dangerous outcomes. Our goal is to move beyond the simple question of "did a goal occur?" and instead quantify danger creation using outcomes like shot quality, first contact, and possession recovery using a complete list of metrics such as ball curl, location, height advantages, and player positioning.

1.3 Existing Research

Existing studies either in the English Premier League or other top-level soccer leagues/competitions have shown similar low conversion of corners, as low as 1.3% (Wei et al., 2025). General trends show that an inswinging cross, where the ball spins towards the goal, leads to more dangerous chances, and Premier League players tend to crowd the goalkeeper more than other leagues.

The majority of these studies only look at goals, and do not give credit to shots created from corners, nor do they register the possession after the corner. These outcomes are far more common and also meaningful for evaluating broader corner quality. Some studies highlight individual players with great success from corners, but this is not useful for teams developing a general strategy of studying the location, curve, and spacing of players for a corner. These gaps motivated our dataset to capture dangerous outcomes such as shots and goals without excluding possession recovery and passive outcomes to better evaluate the full effectiveness of corners.

1.4 Data Preview

Premier League Corners

Attack	Defense	Score	Time	A.Height	D.Height	Goal.Height	Left.Side	Inswinger
Tottenham	Everton	0	19	184.0	182.7	193	0	1
Everton	Tottenham	-1	24	182.5	184.2	NA	0	1
Tottenham	Everton	1	33	184.0	182.7	NA	0	1
Tottenham	Everton	1	34	184.0	182.7	NA	0	1
Everton	Tottenham	-1	44	182.5	184.2	NA	1	1

A.in.6	A.in.Box	D.in.6	D.in.Box	First.Contact	Outcome	Possession	Location	Short
3	5	6	9	1	Goal	1	Back Post	0
4	5	7	9	1	Offsides	1	Front Post	0
4	5	7	9	0	Corner	1	Goalmouth	0
5	6	8	9	0	Shot off Target	1	Goalmouth	0
5	6	8	9	0	Opponent Possession	1	Goalmouth	0

Table 1. displays a complete preview of 5 columns of our dataset, from a match between Everton and Tottenham.

1.5 Data Collection

Our data was collected from the 2025/26 English Premier League Season. Each member of our team selected up to five games at random from matchweeks 7-11 and intently watched at least 50 corners from the games from that designated period, totaling 256 distinct corner kicks, from October 5 to November 9, 2025. All observers used the same coding sheet for consistency. Each observation represents a unique corner taken this season, excluding only corners taken by winning teams later than 90 minutes into the game, as these tend to be taken short to waste time with no effort to get a chance at a goal.

Variable Definitions:

Attack — Team taking the corner kick.

Defense — Team defending the corner.

Score — Attacking goals minus defending goals at time of the corner.

Time — Minute of match when the corner was taken.

A.Height — Average height (cm) of attacking outfield players.

D.Height — Average height (cm) of all defending players.

Goal.Height — Height of the goalscorer if a goal was scored

Left.Side — 1 = left corner flag, 0 = right.

Inswinger — 1 = ball curls toward goal, 0 = away.

A.in.6 — Number of attacking players in the 6-yard box.

A.in.Box — Attacking players in the penalty area.

D.in.6 — Defending players in the 6-yard box.

D.in.Box — Defending players in the penalty area.

First Contact — Which team made the first contact after the corner was taken (1 = Attack)

Outcome — Event resulting from the corner (Goal, Shot, etc.).

Possession — Which team first recovered secure possession (1 = Attack).

Location — Target area of cross (Front Post, Back Post, etc.).

Short — Whether the corner was played short (1 = yes).

2. Summary of the Data

2.1 Summary Tables

Categorical Variable Summary

Frequencies and Relative Percentages

		Count	%	Relative_freq
First.Contact	0	154.0	60.2	
	1	101.0	39.5	
	NA	1.0	0.4	
Inswinger	1	188.0	73.4	
	0	43.0	16.8	
	NA	25.0	9.8	
Left.Side	0	134.0	52.3	
	1	122.0	47.7	
Location	Goalmouth	76.0	29.7	
	Front Post	57.0	22.3	
	Back Post	46.0	18.0	
	Penalty Spot	45.0	17.6	
	Short	25.0	9.8	
	Top of Box	6.0	2.3	
	Other	1.0	0.4	
Outcome	Cleared	94.0	36.7	
	Opponent Possession	38.0	14.8	
	Shot off Target	28.0	10.9	
	Corner	18.0	7.0	
	Goal	16.0	6.2	
	Shot Blocked	16.0	6.2	
	Foul	15.0	5.9	
	Counter	12.0	4.7	
	Shot on Target	12.0	4.7	
	Offsides	6.0	2.3	
	Penalty	1.0	0.4	
Possession	1	156.0	60.9	
	0	95.0	37.1	
	NA	5.0	2.0	
Short	0	225.0	87.9	
	1	31.0	12.1	

Table 2. displays counts, percentages, and relative frequency bars for each categorical variable we recorded. *First Contact* tracks if the attacking team makes the first touch on the ball after the

corner and the rest of the binary variables (*Inswinger*, *Left Side*, *Possession*, and *Short*) were all coded as binary indicators. Locations were demarcated as six zones of the field with respect to the ball's proximity to the goal, front or back post, and boxes. For outcomes, the play was observed until stopped by one of eleven possible outcomes that tracked if possession was kept, a shot was made, or any stoppages made by referees. The most common outcome was *Cleared* followed by *Opponent Possession* and *Shot off Target*; only 6.2% of recorded corners directly resulted in a goal. Corners were taken from both sides of the field at similar rates but a majority were inswingers and the most common locations were the goalmouth and front and back posts. Interestingly, first contact was dominated by the defending team but the attacking team maintained possession for approximately the same proportion of plays. 12.1% of players in the data set kicked short corners, possibly indicating attacking teams use corners to generate offensive pressure or momentum. Overall, in our Premier League dataset corner kicks create quick defensive action and generally move the ball goalwards but plays more often ended in turnovers or clearances than scoring opportunities.

Numerical Variable Summary

Minimum, Maximum, Mean, Median, Standard Deviation, Skewness

	Min	Max	Mean	Median	SD	Skew
A.in.6	0.00	7.00	2.75	3.00	1.76	0.15
A.in.Box	2.00	9.00	6.09	6.00	1.01	−0.56
D.in.6	0.00	9.00	5.43	6.00	2.04	−0.71
D.in.Box	1.00	10.00	8.86	9.00	1.42	−2.71
Score	−3.00	3.00	−0.25	0.00	1.32	−0.04

Table 3. presents the minimum, maximum, mean, median, standard deviation, and skewness for each of the quantitative corner-kick variables. *A.in.6* and *A.in.Box* measure the number of attacking players in the six-yard box and penalty area, respectively, while *D.in.6* and *D.in.Box* record the corresponding number of defenders in those zones. *Score* reflects the goal differential at the time of the corner, measured relative to the attacking team. On average, teams were utilizing more players in the six-yard box and the penalty box when defending against a corner kick, though defending teams also show increased variation in their spatial placement of players, demonstrated by higher standard deviations. Skewness values further reveal that defenders tend to have more players in the box at the time of the corner. The score variable centers around zero, most of our corner kicks took place during tied or near-tied games. This table suggests that defending teams tend to crowd the box more than the attacking team during corner kicks, and that they can occur at high score disparities of up to three goals but center around tied games.

2.2 Figures

Count of Corners Played Heatmap (Figure 1)

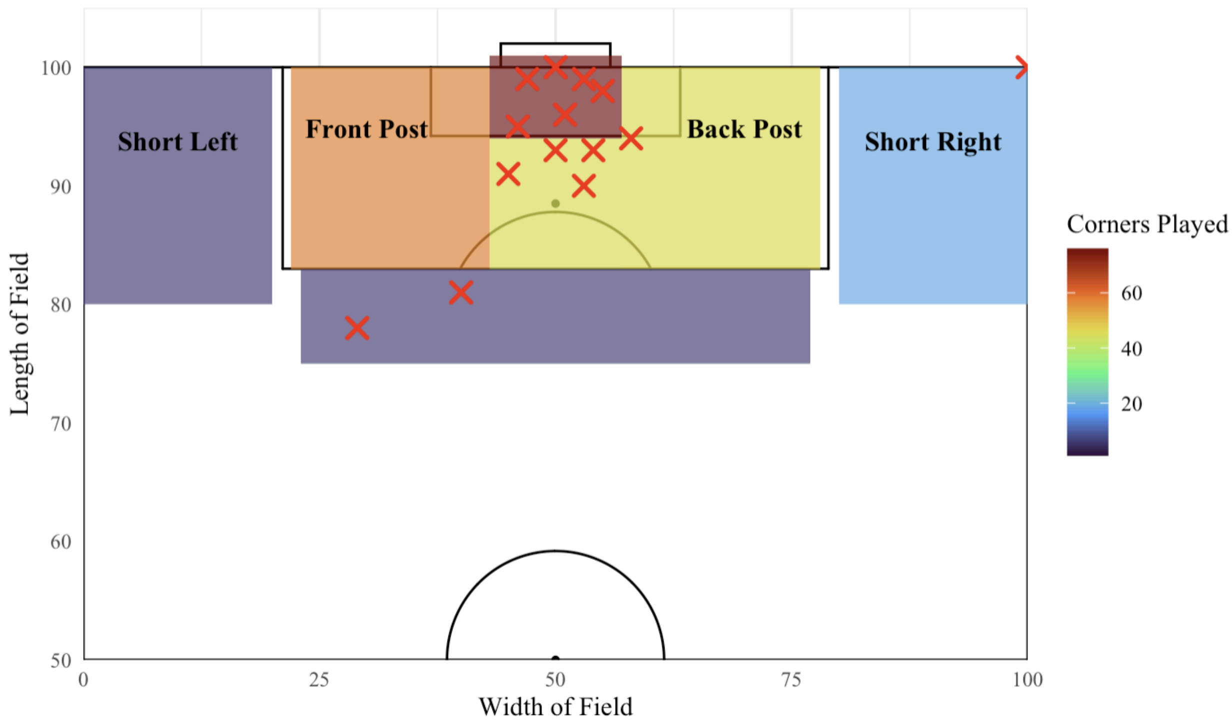

Figure 1 is a heat map created to illustrate the count of corners played into each location on the field, as well as the locations of players for each goal recorded. Each zone represents an area of the field recorded in our "Location" variable. Short passes were split based on whether they occurred on the right or left side of the field. In this figure, the left side of the field and front post are represented on the left half, while the right side of the field and back post are represented on the right half. It's important to note that we did not split the front and back posts by side of the field, and the counts you see are the totals for the front/back post, regardless of side. Each red X-mark represents the location of a goalscorer from where he kicked the ball into the goal. For example, one goal was scored from the corner kick itself; thus, the X-mark is in the corner from which he scored.

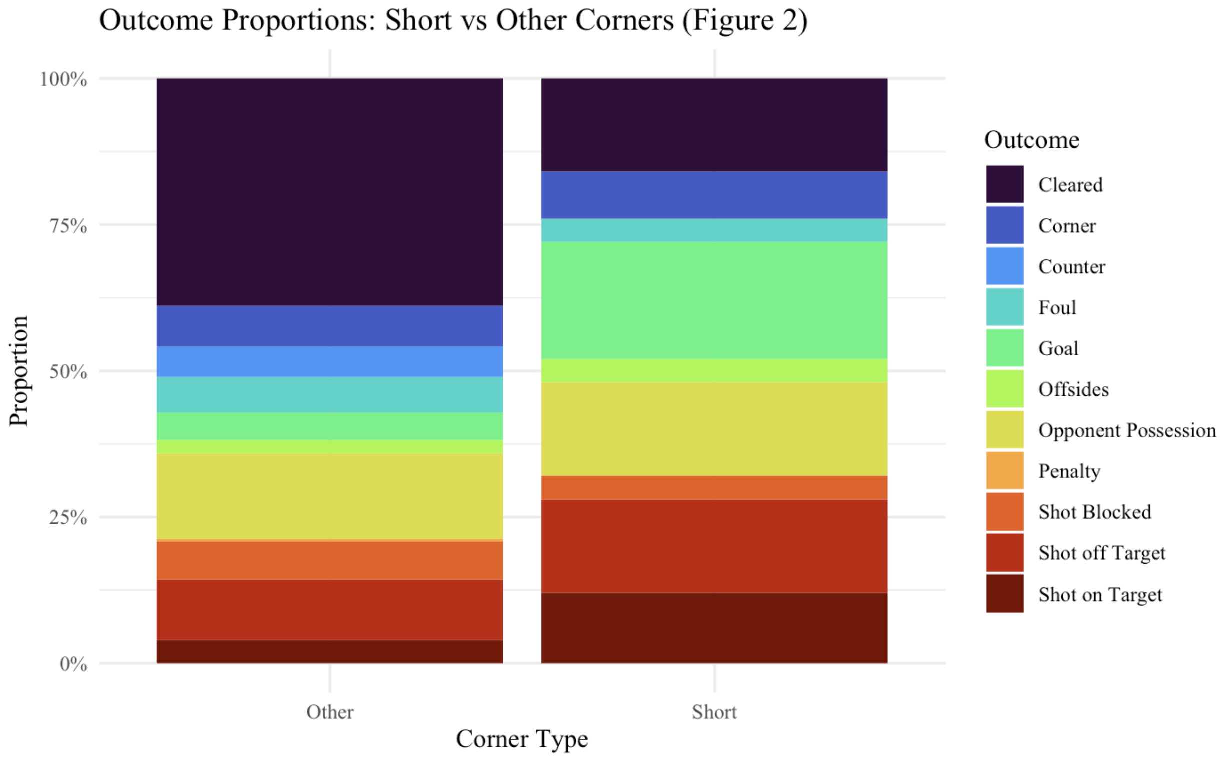

Figure 2 is a stacked bar chart illustrating the difference in proportions of outcomes for short corners compared to all other locations of corners. Each color of the bar represents the proportion of a specific outcome, with the right representing short corners and the left representing all others. Proportions were calculated by dividing the count for a specific outcome, given that it is either "Other" or "Short," by the total counts of "Other" or "Short," depending on the type of corner. These are not the proportions relative to **ALL** corners in the dataset.

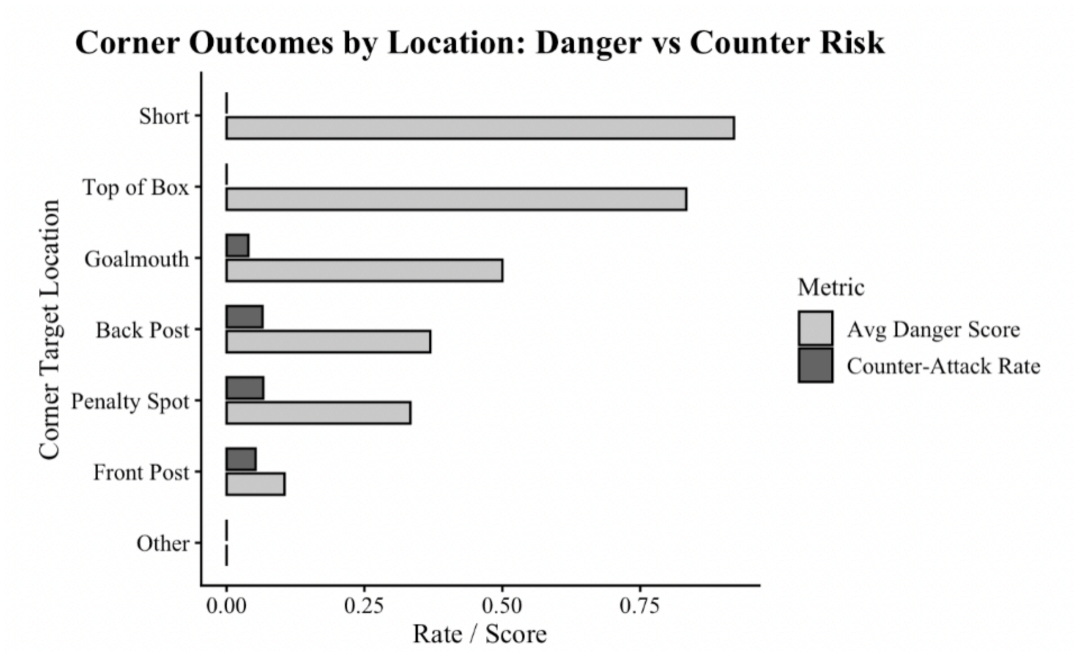

Figure 3 displays the average attacking danger and counter-attack frequency for each recorded corner target location. Attacking success was measured using a weighted danger ranking system we developed that assigns a numerical threat value to each observed corner outcome based on its probability of producing a goal. To break down the system, goals were assigned a value of 3, shots on target and blocked shots were assigned a value of 2, shots off target were assigned a value of 1, and all other outcomes were assigned a value of 0. This approach captures the level of danger on a continuous scale. Finally, the Counter-Attack rate represents the frequency of counterattacks by the opponent for each of the Target Locations

3. Insights from the Data

As shown in Table 2, the majority of corner kicks in the dataset resulted in passive outcomes such as defensive clearances or opponents' possession, with only 6.2% of all corners directly producing a goal. Additionally, only 12.1% of all recorded corners were taken short, with the remaining corners classified as “other” corners, representing traditionally delivered crosses. Observing Figure 2, we can see a clear relationship that short corners produced a substantially higher proportion of shots on target, shots blocked, and goals compared to traditionally delivered corners. Furthermore, other corners resulted in more clearances and opponents' possessions. Short corners produced a danger rate of 35.5% compared to the 14.7% for other corners, and the average danger was almost three times higher for short corners, as short corners had an average of .9 while other corners averaged .34 danger score. A chi-square test confirmed that the difference between short and other corners is statistically significant with a p-value .0086. A two-sample t-test on the danger score showed short corners generate significantly higher quality attacking outcomes on average with a p-value of .022.

This pattern was confirmed in Figure 3, which directly compares the average danger score and counter-attack rate by where the set piece was placed, our corner target location variable. The most dangerous play for the attacking team was short corner routines with an average danger score of .92, followed by the top of the box with .83. The top-of-the-box result must be interpreted with caution with its small sample size (n = 2). These locations both produce higher danger rates than the traditional crossing placement used on corners, such as the goalmouth (.50), backpost (.37) and penalty spot (.33). Corners aimed at the front post (.11) and the miscellaneous “other” locations produced a very little danger threat. This demonstrates that attacking danger varies by different target locations, with short corner routines producing the highest average quality of chances. The penalty spot recorded the highest rate of counter-attack at 6.7%, followed by the back post at 6.5%. Even though short corners produced far more attacking danger, they did not make teams more vulnerable in transition. Short corner routines give the best balance of risk and reward, as it increases threat without raising the chance for an immediate counter. Tactically, coaches can view this to boost corner efficiency while keeping their teams defensive stability intact.

A common point stressed by many coaches from around the globe surrounding corners is that winning first contact determines who would maintain possession. As seen in Table 2, we found that the defending team made first contact 60.2% of the time, but in spite of that the attacking team recovered possession 60.9% of the time. Suggesting an altering proposition: that winning the first contact on a corner has no effect on whether the attacking team retains possession. To further examine this insight, we conducted a chi-square test of independence between first contact and possession, which resulted in a p-value of 0.994, indicating no statistical relationship between the variables at all. Which divulges that defensive first touches frequently result in partial clearances or uncontrolled headers that attackers are able to immediately challenge. As possession is decided less by initial contact, but more by the structure and spacing of the players that are ready for the second chance opportunity.

We also then compared danger scores between corners where the attacking team won first contact and those where the defense did using the weighted danger scale from earlier. A two-sample t-test provided a p-value of 0.0000098, which showed a significant difference. When this result is viewed alongside Figure 3 (indicates that some target locations naturally produce more danger than the other), it shows that attacking first contacts tend to happen in these high-value zones (like the goalmouth) where most touches lead to a chance. However, defensive first contacts often happen in safer areas or result in clearances, so they naturally produce lower danger scores. Which showcases that the delivery and location of the corner are doing the majority of the work and not actually winning the first touch. This aims us towards our insight that teams would get more out their corners if they planned for second balls and shaping the space around the penalty area, instead of focusing on winning the initial first contact.

If we were to repeat this study, we would make several changes to our data collection and design. We would observe marking schemes more carefully, since whether a defense uses a zonal or man marking is likely a confounding variable that influences both how short corners are defended and how dangerous they become. Our current dataset treats all defensive structures as equal, which makes it hard to separate how defenses game-planned the set piece. We would also record the height of the attackers involved in each corner, rather than just the average height of each team's lineup. Additionally, we would want to take account for the locations of the attackers and defenders inside of the box in order to track which setups led to more success or failure. As well as the location of second chance balls in order to see if there are any high frequency areas and how to plan for the second chance opportunities. Finally, because some of our investigations had small sample sizes, especially with certain locations of corners, we would expand data collection across more matches or seasons to support stronger tests and more confident recommendations for corner kick decisions.

References

Ali, T. (2023, August 9). *Premier League Corners: Inswingers vs Outswingers, Short vs Crossed & More*. Opta Analyst. Retrieved December 3, 2025, from https://theanalyst.com/articles/premier-league-corner-stats-inswingers-vs-outswingers-short-vs-crossed-kai-havertzs-defending

Creative Set Plays. (2025, August 5). *Premier League 2025–26: Set Play Coaches*. Creative Set Plays. Retrieved December 3, 2025, from https://www.creativesetplays.com/post/premier-league-set-piece-coaches-2025-26

Plakias, S., Armatas, V., & Giakas, G. (2025, April 30). Technical–Tactical Analysis of Corner Kicks in Male Soccer: A Systematic Review. *Applied Sciences*, *15*(9). MDPI. Retrieved December 3, 2025, from https://www.mdpi.com/2076-3417/15/9/4984

Premier League. (2025, May 25). *The 2024/25 Premier League season in numbers*. Premier League. Retrieved December 3, 2025, from https://www.premierleague.com/en/news/4316617

Soccer.com. (2025, March 27). *The Basic Rules of Soccer: A Complete Guide*. Soccer.com. Retrieved December 3, 2025, from https://www.soccer.com/guide/rules-of-soccer-guide

Wei, X., Tang, C., Chmura, P., Krustup, P., & Bjørknæs, L. (2025, July 1). Understanding the variability and determinants of corner kick effectiveness in professional football: A comparative and predictive analysis. *Intelligent Sports and Health*, *1*(3), 179-185. https://www.sciencedirect.com/science/article/pii/S3050544525000337